

Pwoka Technologies Ltd.

Enterprise Network Design and Deployment Progress Report

1. Executive Summary

Project Title: Multi-Site Enterprise Network Infrastructure Design

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Organization: Pwoka Technologies Ltd.

Objective: Design, implement, secure, and test a highly available, scalable enterprise network infrastructure that successfully connects a multi-department Head Office campus to a geographically remote Branch Office.

This report summarizes the work completed during the design and implementation stages of the project. The network has been developed in Cisco Packet Tracer using Cisco enterprise devices and follows a hierarchical network design. The current implementation includes the physical topology, VLAN segmentation, WAN connectivity, inter-VLAN routing, dynamic routing, gateway redundancy, and Layer 2 security features.

2. Network Topology Overview

The enterprise network consists of a Head Office and a Branch Office connected through point-to-point WAN links.

The Head Office contains two Cisco 2911 routers, two Cisco Catalyst 3560 multilayer switches, and two Cisco Catalyst 2960 access switches. These devices provide redundancy, routing, and user connectivity.

The Branch Office consists of one Cisco 2911 router and one Cisco Catalyst 2960 access switch serving the remote users.

The hierarchical design was selected because it is easier to manage, supports future expansion, and allows redundancy features such as HSRP and EtherChannel to be implemented.

3. Hardware Used

The following Cisco devices were used during the implementation:

- Cisco 2911 ISR Routers (3)
 - HQ-R1
 - HQ-R2
 - BR-R1
- Cisco Catalyst 3560-24PS Multilayer Switches (2)
 - HQ-DSW1
 - HQ-DSW2
- Cisco Catalyst 2960-24TT Access Switches (3)
 - HQ-ASW1
 - HQ-ASW2
 - BR-ASW1

End devices include:

- 11 PCs
- 2 Network Printers
- 1 Server

4. VLAN Design and IP Addressing

The network is divided into separate VLANs to improve security, reduce broadcast traffic, and simplify network management.

Head Office

- VLAN 10 – Sales – 192.168.10.0/24
- VLAN 20 – IT – 192.168.20.0/24
- VLAN 30 – Printers – 192.168.30.0/24
- VLAN 40 – Human Resource – 192.168.40.0/24
- VLAN 50 – Management – 192.168.50.0/24

VLAN 100 – Server Network – 192.168.100.0/24

Branch Office

VLAN 30 – Printers – 10.10.30.0/24

VLAN 40 – Users – 10.10.40.0/24

All LANs use a /24 subnet mask, while WAN links use /30 subnet masks to efficiently utilize IP addresses.

5. WAN Design

Two serial WAN links connect the enterprise network.

HQ-R1 to HQ-R2 using network 172.16.12.0/30

HQ-R2 to BR-R1 using network 172.16.23.0/30

The DCE interfaces provide clocking for the serial links. These connections allow communication between the Head Office and the Branch Office.

6. Physical Topology and Cabling

The network topology was created in Cisco Packet Tracer following the planned design. Copper straight-through cables were used between routers and switches, while the connection between the distribution switches was prepared for EtherChannel operation. Serial DCE and DTE cables were used for the WAN connections between routers. All devices were renamed using meaningful hostnames to simplify administration and troubleshooting.

7. Basic Device Configuration

Initial configuration was completed on all routers and switches.

The following settings were applied:

Device hostnames

- Enable secret passwords
- Console passwords
- Password encryption
- Login banner
- Local administrator account
- Domain name
- SSH Version 2 for secure remote management

These settings improve the security of the management plane and replace insecure default configurations.

8. Inter-VLAN Routing and EtherChannel

Inter-VLAN routing was configured using Switch Virtual Interfaces (SVIs) on the multilayer switches. IP routing was enabled to allow communication between different VLANs. An LACP EtherChannel was created between the distribution switches to increase bandwidth and provide link redundancy between the core switching devices.

9. Dynamic Routing

EIGRP Autonomous System 100 was configured on all routers. The routing protocol advertises the Head Office, Branch Office, and WAN networks, allowing routes to be exchanged automatically. Automatic summarization was disabled using the **no auto-summary** command to support classless routing.

10. Final implementation progress

Following the completion of the network design phase, all planned configurations were successfully implemented and tested in Cisco Packet Tracer. The project progressed from a basic network topology to a fully operational enterprise network connecting the Head Office and Branch Office.

10.1 Dynamic Routing (EIGRP)

EIGRP Autonomous System (AS) 100 was configured on all routers and multilayer switches to provide automatic route exchange throughout the network. The **no auto-**

summary command was applied to support the different subnet sizes used in the project.

Verification

EIGRP neighbours formed successfully.

Routing tables updated dynamically.

Communication between the Head Office and Branch Office was verified.

10.2 Gateway Redundancy (HSRP)

HSRP was configured on both multilayer switches to provide gateway redundancy for all Head Office VLANs. HQ-DSW1 was configured as the Active gateway for VLANs 10, 20 and 30, while HQ-DSW2 became the Active gateway for VLANs 40, 50 and 100. Both switches were configured with **preempt**, allowing the preferred gateway to automatically resume control after recovery. The virtual gateway address ending in **.254** was assigned as the default gateway for all Head Office VLANs.

Verification

Active and Standby roles formed successfully.

Gateway failover operated correctly during testing.

10.3 EtherChannel and Trunk Configuration

An EtherChannel was created between HQ-DSW1 and HQ-DSW2 using LACP. The bundled interfaces were configured as an IEEE 802.1Q trunk to carry all required VLANs between the distribution switches.

Verification

Port-Channel 1 became operational.

Trunk links carried all configured VLANs successfully.

10.4 Layer 2 Security

To improve access-layer security, the following features were implemented on the access switches:

- Port Security using Sticky MAC addresses with a maximum of one device per port.

- DHCP Snooping to prevent rogue DHCP servers.

- Port Fast enabled on end-user interfaces for faster network access.

These configurations improve security while maintaining normal network operation.

10.5 Device Management Security

All network devices were configured with secure management settings including:

- Device hostnames

- Enable secret password

- Console and VTY passwords

- Password encryption

- Local administrator account

- SSH Version 2

- Login banner

These settings help protect network devices from unauthorized access.

11. Network Testing and Verification

After configuration, several tests were performed to confirm that the network operates correctly.

The following commands were used during verification:

- show Ip route

- show Ip eigrp neighbours

- show standby brief

- show EtherChannel summary

```
show interfaces trunk
show port-security interface
show ip dhcp snooping
```

Successful testing confirmed:

- Dynamic routing is operational.
- HSRP redundancy is functioning correctly.
- EtherChannel is active.
- Trunk links are carrying the required VLANs.
- Security features are working as expected.
- End-to-end communication between the Head Office and Branch Office is successful.

12. Final Project Status

The enterprise network has been fully implemented and tested according to the project requirements.

The completed network includes

- Multi-site WAN connectivity
- VLAN segmentation
- Inter-VLAN routing
- Dynamic routing using EIGRP
- Gateway redundancy using HSRP
- EtherChannel link aggregation
- SSH secure management
- Port Security
- DHCP Snooping
- Port Fast

All configured services were verified in Cisco Packet Tracer and the network is operating as expected.

13. Conclusion

The Pwoka Technologies Ltd. enterprise network project was successfully completed in Cisco Packet Tracer. The project demonstrates the practical application of CCNA concepts including VLAN implementation, multilayer switching, WAN connectivity, dynamic routing, gateway redundancy, Layer 2 security and secure device management. The completed network is stable, secure and scalable, providing reliable communication between the Head Office and Branch Office while meeting the objectives of the project.